Findings: Implicit VI on a Banana Distribution

Problem. Approximate a two-dimensional banana-shaped distribution with Gaussian VI, adversarial density-ratio estimation (DRE), and semi-implicit VI (SIVI). The target is $\log \tilde{p}(z) = -z_1^2/2 - (z_2 - z_1^2)^2/(2 \times 0.3^2)$. Exact target samples are used only for evaluation. Each method has 3,000 VI updates with batches of 512 and seed 7.

Gaussian baseline. VI maximizes the ELBO, $L = E_q[\log \tilde{p}(z) - \log q(z)]$: target fit plus entropy. The full-covariance Gaussian has analytic entropy, so its objective is tractable. It can rotate and stretch an ellipse but cannot follow a curved ridge.

Adversarial DRE. A neural generator maps normal noise to samples, but $\log q$ is unknown. A balanced classifier distinguishes generator samples (label 1) from reference samples $r = N(0, 9I)$ (label 0). Its ideal logit is $T = \log(q/r)$, so $\log q \approx T + \log r$. The generator minimizes $E[T + \log r - \log \tilde{p}]$. Five classifier updates alternate with one generator update. Generator samples are detached for classifier training; classifier weights are frozen while input gradients remain active for generator training. Imperfect ratios bias the update; a fast-moving generator can outrun the classifier.

SIVI bonus. A neural network generates random means for explicit Gaussian conditionals. The unknown marginal density is replaced by a 25-component mixture, including the conditional that generated each sample. This gives a lower ELBO surrogate in expectation. More components tighten it under regularity assumptions but increase computation. No classifier or alternating game is needed.

Method	MMD^2 ↓	Training (s)
Gaussian	0.1375	4.9
DRE	0.0011	43.2
SIVI	0.0230	10.9

MMD^2 compares fresh samples with the target at kernel bandwidth 1; lower is better.

Findings. DRE had the lowest MMD^2 in this run. DRE followed the curved ridge more closely than the Gaussian. Gaussian VI is fast but shape-limited; DRE is flexible but sensitive to classifier quality; SIVI avoids that instability but has a finite-mixture gap. These single-seed results do not establish a universal winner.